Ethan Theodore Xiong

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RESEARCH OUTPUTS

Conference Posters

- **Xiong, E. T.**, Sapkota, N., Tordiff, J., Rogak, S. N., Corbin, J. C., and Sipkens, T. A. "Image analysis techniques for transmission electron microscopy analysis of carbonaceous and related particles and their application." Canadian Aerosol Conference, 2026.

Publications

- **Xiong, E. T.**, Sapkota, N., Sipkens, T. A., and Rogak, S. N. "An improved automated deep-learning workflow for segmenting and characterizing carbonaceous particles in transmission electron microscopy images." Manuscript in preparation; target journal: Powder Technology

- Sapkota, N., Mehri, R., Corbin, J. C., **Xiong, E. T.**, Rogak, S. N., and Sipkens, T. A. "Investigating morphology of fresh and compacted soot aggregates using bidimensional effective density and transmission electron microscopy." Manuscript in preparation; target journal: Journal of Aerosol Science.

- Sharma, S., Underhill, T., **Xiong, E. T.**, Sapkota, N., Lintusaari, H., Kitchen, P., Goeing, J., Rogak, S. N. "Integrated Screening of Alternative Aviation Fuels: Development and Commissioning of a Laboratory Platform for Engine Performance and Emissions Characterization." Manuscript in Preparation; target journal: Aerospace Science and Technology

EDUCATION

University of British Columbia, Vancouver, BC **September 2021 - May 2027**

Bachelor of Applied Science in **Mechanical Engineering**, Specialization in **Aerospace** (Co-op Option)

- GPA: 3.70/4.33
- Key Courses: Turbulent Shear Flow (91%), Computational Fluid Dynamics (93%), Finite Element Analysis (92%), Experimental Thermofluids (89%), Heat Transfer (100%)

RESEARCH INTEREST

- **Gas-Turbine Combustion and Turbomachinery Performance:** Investigating how combustor design, operating conditions, and fuel properties influence efficiency, operability, flame stability, heat transfer, pressure losses, and overall propulsion performance.
- **Sustainable Aviation Fuels, Soot Formation, and Combustion Emissions:** Studying how conventional and alternative jet fuels affect soot/nvPM formation, gaseous emissions, particle morphology, and fuel-specific combustion behaviour in propulsion-relevant systems.
- **Turbulent Reacting Flows, CFD, Turbulence Modelling, and Aerosol Diagnostics:** Applying computational and/or experimental methods to study turbulent mixing, particle-laden flows, particle sampling, and deep-learning-based image processing.

RESEARCH EXPERIENCES

Undergraduate Research Assistant, *UBC Energy & Aerosol Laboratory,* Vancouver, BC **May 2026 – August 2026**

Supervisor: **Prof. Steven Rogak**

- Developed a deep learning workflow to detect, segment, and measure the morphology of soot and other carbonaceous particles in TEM images, reducing analysis time by 99.6% compared with manual analysis; manuscript in preparation.
- Led the redesign of a thermophoretic TEM sampler for field aerosol collection, using Ansys Fluent to evaluate jet temperature gradients, nozzle velocity, and downstream thermal behaviour for improved particle deposition.
- Designed a gusseted support-frame concept for a mobile jet-engine exhaust duct system with a ~12 m vertical riser, using CAD and FEA for the support structure and MATLAB air-entrainment and heat-transfer modelling to assess flow capacity, mixed-gas temperatures, and duct sizing.

PROFESSIONAL EXPERIENCES

Junior Designer – Co–op, *Smith and Andersen,* Burnaby, BC **January 2025 – August 2025**

- Delivered mechanical designs across 10+ projects, incl. a 3-tower residential complex and a commercial pool facility (Burnaby), producing coordinated plumbing, drainage, hydronic, domestic water, sanitary & HVAC layouts.
- Developed models, calculations, and documentation using Revit, AutoCAD, Excel, Bluebeam Revu, and Elite CHVAC, ensuring compliance with the BC Plumbing Code and ASHRAE 62.1.

- Verified pool heat exchanger performance under pool-fill conditions by developing an iterative LMTD-based heat transfer calculator, identifying significant discrepancies with manufacturer specifications and recommending vendor validation to mitigate design risk.

Mechanical Engineering Intern – Co-op, *Ausenco,* Burnaby, BC **September 2023 – January 2024**
- Developed an MS Excel VBA tool to design a step shaft for a pivoting chute in compliance with AS-1403, integrating bearing selection, load calculations, and reaction force analysis to validate structural integrity.
- Led the development of an idle time forecasting tool utilizing Excel VBA, decreasing manual analysis time by 20%, saving approximately 2 hours per week.
- Implemented FEA and DEM simulations using ANSYS to validate chute designs, reducing design iteration time by 30% and ensuring structural integrity under various load conditions. Enhanced design feasibility by identifying potential failure points early, contributing to a 15% reduction in material costs through optimized design.

Manufacturing Engineering - Co-op, *SSAB,* Delta, BC **May 2023 – September 2023**
- Utilized Autodesk ReCAP point cloud modelling and Inventor to assemble bucket liners with 98% precision, optimizing the manufacturing process.
- Prepared 100+ CNC-ready nests for plasma and oxy-fuel operations using SigmaNEST, adjusting layouts with operators when actual plate conditions differed from system inputs to minimize downtime.

SELECTED TECHNICAL PROJECTS

Large-Eddy Simulation Subgrid-Scale Model Review, MECH 586: Turbulent Shear Flow **January 2026 – April 2026**
- Conducted a technical literature review of eddy-viscosity closure models for large-eddy simulation, comparing Smagorinsky, dynamic Smagorinsky, WALE, Vreman, and Sigma formulations.
- Evaluated model assumptions, near-wall scaling behaviour, strain- and rotation-rate sensitivity, and suitability for wall-bounded turbulent shear flows, with emphasis on the WALE model.

Smart Brake Chamber, *Level 5 Fleet CAPSTONE Project,* Vancouver, BC **September 2025 – April 2026**
- Validated project feasibility using rapid prototyping by designing an automatic locking mechanism, producing 10+ iterative SolidWorks models, and fabricating multiple 3D-printed components integrated with motors and actuators, successfully demonstrating 100% functional performance in testing.
- Created complete drawing sets for the invention, including exploded assembly views and professional engineering drawings for all custom parts, while performing tolerance stack-up analysis and FEA to verify manufacturability, fit, and structural integrity of critical components.

Zero-Dimensional Cycle Analysis of a Mixed Turbofan Aero Engine, **September 2025 – December 2025**
MECH 477: Aerospace Propulsion
- Developed a zero-dimensional MATLAB cycle model for a mixed-flow turbofan engine, applying station-by-station mass, energy, and momentum balances across major engine components.
- Performed parametric studies to evaluate how bypass ratio, compressor pressure ratio, turbine inlet temperature, and component efficiencies affect thrust, specific fuel consumption, and overall propulsion performance.

STOODIS (Solid Motor Rocket), *Thunderbird Aerospace,* Vancouver, BC. **September 2023 - September 2025**
- Designed and manufactured rocket components, including aluminum fuel tank features and a von Kármán-profile nosecone, using SolidWorks CAD, lathe machining, and precision surface finishing to meet tight dimensional tolerance requirements.
- Performed heat transfer analysis near the nozzle region of a liquid-propellant rocket to estimate thermal loading and support selection of appropriate ablative liner / thermal protection thickness for critical components.

KEY COMPETENCIES

Scientific Computing & Data Analysis: Python, MATLAB, NumPy, OpenCV, Matplotlib, Excel VBA, LaTeX
CFD, FEA & Modelling: ANSYS Fluent, OpenFOAM, ANSYS Mechanical, Abaqus, SolidWorks, Onshape
Experimental & Instrumentation: Arduino, ESP32, thermocouples, pressure/flow sensors, oscilloscope, data logging
Manufacturing & Prototyping: Lathe, mill, 3-axis CNC, 3D printing, rapid prototyping

LEADERSHIP / TEACHING

Research Mentorship – TEM Image Processing Project **July 2026 – August 2026**
- Supervision of Grace Kirchen and Elliot Damgaard in manual segmentation and data analysis for TEM images of soot and related carbonaceous particles.
- Coordinate annotation tasks, review segmentation consistency, and provide guidance on particle identification, mask preparation, and validation data quality for deep-learning model evaluation.